

SOFTWARE REQUIREMENTS SPECIFICATION

TechLearn LMS

Enterprise Tech Training Platform — Next Generation

Version 2.0 — Refined & Extended

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Classification: Confidential

Changelog: v2.0 incorporates competitive differentiators, learner accountability systems, anti-cheating enforcement architecture, and Nigeria-specific market requirements.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) v2.0 defines the complete functional and non-functional requirements for TechLearn LMS — an enterprise-grade, web-based Learning Management System purpose-built for technical training. This version incorporates a competitive differentiation layer, a comprehensive learner accountability and academic integrity framework, and regional market adaptations (with particular focus on Nigeria and emerging markets). It serves as the authoritative reference for product managers, architects, developers, QA engineers, and all stakeholders involved in the design, development, and delivery of the platform.

1.2 Scope

TechLearn LMS is an enterprise-scale, cloud-native, browser-based SaaS platform designed to support 50,000+ concurrent users. It delivers structured technical curricula across the following domains:

- Software Development — Frontend, Backend, Full-Stack
- Cybersecurity
- Cloud & DevOps
- IT Support & Networking
- Programming Languages (Python, JavaScript, Java, Go, Rust, SQL, and more)
- Data Analysis & Data Engineering

The system supports four user roles: Students/Learners, Instructors/Trainers, Administrators, and Corporate HR/Managers — each with distinct capabilities and access controls. TechLearn is positioned to compete directly with Pluralsight, Udemy for Business, Coursera for Enterprise, and LinkedIn Learning by offering deeper technical tooling, verifiable outcomes, and superior learner accountability infrastructure.

1.3 Competitive Positioning Summary

The following table identifies where market competitors fall short and how TechLearn addresses each gap:

Competitor Weakness	TechLearn's Answer
Certificates are not employer-trusted	Proctored assessments + cryptographically verifiable badges
No real coding accountability	Keystroke logging IDE, MOSS plagiarism detection, code explain (viva)
Videos counted as watched on play	Active watch detection, comprehension checkpoints, seek restrictions

No career outcome data	Post-completion career tracker + job board integration
Inconsistent course quality (Udemy model)	Curated instructor marketplace with outcome-based quality scoring
Poor enterprise ecosystem integration	First-class REST API, webhooks, Workday/SAP/Salesforce connectors
No cognitive science in retention	Spaced repetition engine based on Ebbinghaus forgetting curve
No team-based technical learning	Cohort hackathon mode with shared IDE + leaderboards
Excludes low-bandwidth markets	PWA offline mode, adaptive streaming, mobile-first IDE, Paystack/Flutterwave

1.4 Definitions & Acronyms

Term / Acronym	Definition
LMS	Learning Management System
SRS	Software Requirements Specification
IDE	Integrated Development Environment
SSO	Single Sign-On
SCORM	Shareable Content Object Reference Model
xAPI	Experience API (Tin Can API)
RBAC	Role-Based Access Control
CDN	Content Delivery Network
SLA	Service Level Agreement
MFA	Multi-Factor Authentication
PWA	Progressive Web App
MOSS	Measure of Software Similarity (plagiarism detection tool)
SRS (retention)	Spaced Repetition System (learning science method)
DRM	Digital Rights Management
HLS/DASH	HTTP Live Streaming / Dynamic Adaptive Streaming over HTTP
PCI-DSS	Payment Card Industry Data Security Standard
GDPR	General Data Protection Regulation
RPO / RTO	Recovery Point Objective / Recovery Time Objective
CSP	Content Security Policy

1.5 References

- IEEE Std 830-1998: Recommended Practice for SRS
- WCAG 2.1 AA — Web Content Accessibility Guidelines
- OWASP Top 10 Security Standard
- SCORM 1.2 / 2004 Standards
- xAPI (Tin Can) Specification
- Open Badges 2.0 Specification (IMS Global)
- GDPR & CCPA Data Privacy Regulations
- MOSS — Measure of Software Similarity (Stanford)
- Ebbinghaus Forgetting Curve Research

2. Overall System Description

2.1 Product Perspective

TechLearn LMS is a greenfield, cloud-native, browser-based SaaS application. It integrates with third-party services including Zoom (live sessions), Slack (notifications), GitHub (code repository reading), Stripe / Paystack / Flutterwave (payments), and AI recommendation and integrity engines. It is architecturally designed as a microservices system capable of independent scaling per functional domain.

2.2 Product Features Overview

Feature Area	Description	Priority	v2 Status
Video Courses & Quizzes	Structured multimedia courses with embedded assessments	High	Existing
Active Video Accountability	Watch detection, seek restrictions, comprehension checkpoints	High	NEW
Live Virtual Classrooms	Zoom-integrated real-time instructor-led sessions	High	Existing
Inline IDE (Restricted)	Browser-based editor with copy-paste enforcement	High	Existing
IDE Keystroke Recorder	Full typing session replay for instructor code review	High	NEW
Code Plagiarism Detection	MOSS + AI signature analysis for submitted code	High	NEW
Code Explain / Viva Mode	Async video explanation of submitted code by learner	High	NEW
Certifications & Badges	Automated certificate issuance and Open Badges 2.0	High	Existing
Live Proctored Assessments	Webcam, screen recording, browser lockdown for exams	High	NEW
Learner Accountability Dashboard	Health score, streak tracking, behavioral flags	High	NEW
GitHub Repo Integration	Read exercise files from repositories within lessons	High	Existing
Progress Tracking & Analytics	Individual and cohort performance dashboards	High	Existing
Spaced Repetition Engine	Forgetting-curve-based concept resurface system	High	NEW
AI Pair Programmer (Tutor)	Socratic AI assistant in IDE — explains, never solves	High	NEW

AI Course Content Generator	AI drafts course outlines, quiz banks, reading lists	High	NEW
Peer Code Review System	Structured rubric-based peer review before submission	High	NEW
Team Hackathon / Challenge Mode	Cohort-based collaborative coding challenges	Medium	NEW
Career Outcome Tracker	Post-cert job board integration, hiring outcome tracking	High	NEW
White-Label Enterprise Portals	Custom subdomain, branding, catalog per enterprise client	High	NEW
Instructor Marketplace	Curated, outcome-scored instructor ecosystem	High	NEW
AI-Powered Recommendations	Personalized learning paths via ML engine	High	Existing
Payment & Subscription Billing	Stripe + Paystack + Flutterwave, enterprise seats	High	Extended
Multi-language / Localization	UI and content translation including RTL	Medium	Existing
Offline Mode (PWA)	Encrypted offline video with sync, mobile-first IDE	High	NEW
Enterprise API & Webhooks	First-class REST API + Workday/SAP/Salesforce connectors	High	NEW
Biometric Identity Verification	Face scan at quiz/exam entry to prevent proxy learning	High	NEW
Public Portfolio Builder	All student project work auto-compiled into public portfolio	Medium	NEW

2.3 User Classes and Characteristics

2.3.1 Students / Learners

The primary consumers of content. They enroll in courses, complete modules, submit code in the inline IDE, participate in live sessions, and earn certificates. They range from beginner to advanced technical professionals across geographies including Nigeria, Africa, and global enterprise employees.

2.3.2 Instructors / Trainers

Content creators and session facilitators. They build course curricula, upload video content, create quizzes, set up GitHub repository links, review keystroke replay logs, review flagged submissions, and host live virtual sessions.

2.3.3 Administrators

Platform administrators responsible for user management, role assignments, system configuration, content moderation, proctoring oversight, and organization-wide analytics.

2.3.4 Corporate HR / Managers

Organizational managers who assign learning paths to employees, monitor team progress, manage subscription seats, review skill-gap reports, and evaluate career outcome data from completed learning programs.

2.4 Operating Environment

- Deployment: Web-based, browser-only (Chrome 100+, Firefox 100+, Safari 15+, Edge 100+)
- Hosting: Cloud-native (AWS / GCP / Azure) with auto-scaling and multi-region support
- Scale: Enterprise — 50,000+ registered users, 10,000+ concurrent sessions
- Offline: PWA with encrypted local storage and background sync
- Availability: 99.9% uptime SLA
- CDN: Global content delivery for video assets with adaptive bitrate (HLS/DASH)
- Regional Markets: Primary — Nigeria; Extended — Africa, Southeast Asia, Latin America

2.5 Assumptions and Dependencies

- GitHub OAuth App credentials provisioned for read-only repository access
- Zoom API credentials provided for live session integration
- Stripe, Paystack, and Flutterwave API keys provisioned for payment processing
- AI recommendation and AI code-analysis engines available via internal or vendor API
- MOSS integration available via institutional license or API wrapper
- Proctoring infrastructure (webcam access, screen recording APIs) available in target browsers
- Users in offline-mode markets have devices capable of running the PWA with local storage

3. Functional Requirements

3.1 User Authentication & Authorization

3.1.1 Authentication

- The system shall support email/password registration and login
- The system shall support SSO via OAuth 2.0 (Google, Microsoft, GitHub)
- The system shall enforce MFA for all accounts; Admin accounts shall require hardware key or authenticator app
- The system shall lock accounts after 5 consecutive failed login attempts
- The system shall enforce single active session per account — concurrent logins shall be rejected
- Device fingerprinting shall track device/browser identity per user; significant device changes during high-stakes assessments shall trigger a security flag
- IP location tracking shall flag significant geographic location changes during proctored assessments

3.1.2 Role-Based Access Control

- Four roles enforced: Student, Instructor, Admin, HR Manager
- Admins shall assign and revoke roles for any user
- HR Managers shall have read-only access to learner progress within their organization

3.2 Course Management

3.2.1 Course Creation

- Instructors shall create courses with title, description, category, skill level, and cover image
- Courses shall be structured into modules and lessons
- Each lesson shall support: video, text/markdown, code exercises, quizzes, and GitHub file embeds
- Courses shall support draft, review, and published states
- Instructors shall have access to an AI Course Content Generator that, given a topic and target skill level, produces a full course outline, module structure, draft quiz question bank, and suggested reading/documentation list
- Instructors shall be able to set prerequisite courses and skill-gate enrollment based on pre-assessment score

3.2.2 Course Enrollment

- Students shall self-enroll in free or purchased courses
- HR Managers shall bulk-enroll employees into specific learning paths
- The system shall send enrollment confirmations via email and Slack

3.3 Video Learning Accountability

Standard LMS platforms count a video as completed the moment it starts. TechLearn enforces genuine engagement through the following mechanisms:

3.3.1 Active Watch Detection

- The system shall track tab focus, mouse/keyboard activity, and scroll position during video playback
- If a student switches tabs during video playback, the video shall automatically pause and the event shall be logged with a timestamp
- Progress shall only accumulate during periods of verified active engagement; idle or unfocused time shall not count toward completion

3.3.2 Video Comprehension Checkpoints

- Every 5–10 minutes of a lesson video (configurable by instructor), the system shall pause playback and present an unannounced micro-quiz question drawn randomly from a question bank
- The student must answer the checkpoint question correctly before video playback resumes
- Questions shall be randomized per student from a sufficiently large pool to make answer-sharing ineffective
- Checkpoint failures shall be logged and visible in the instructor's accountability dashboard

3.3.3 Timestamped Note Requirements

- Instructors may configure specific video timestamps at which students are required to write a summary note of what was just explained
- Notes shall be tied to the video timestamp and stored for instructor review as evidence of engagement

3.3.4 First-Watch Seek Restriction

- On a student's first viewing of any lesson video, forward-seeking (skipping ahead) shall be disabled
- On subsequent viewings, full seek controls shall be available to support review and revision

3.4 Inline IDE & Code Integrity

The inline IDE is a critical differentiator. It combines a powerful browser-based editor with the most comprehensive academic integrity controls of any LMS on the market.

3.4.1 Editor Capabilities

- The IDE shall support syntax highlighting for: Python, JavaScript, TypeScript, Java, Go, SQL, Bash, HTML, CSS
- The IDE shall support in-browser code execution via sandboxed, network-isolated runtimes
- The IDE shall support multi-file editing for project-based exercises

- Real-time output and error messages shall be displayed in a split-panel terminal view
- Test case validation shall run learner code against hidden instructor-defined test suites

3.4.2 Copy-Paste Restriction

- The IDE shall intercept and block all paste events (keyboard shortcuts, right-click, drag-and-drop)
- The clipboard API (navigator.clipboard) shall be restricted within the editor context via CSP
- All blocked paste attempts shall be logged with timestamp and surfaced in the instructor dashboard
- The student shall see a clear, non-punitive warning message on each blocked paste attempt
- IDE-native autocomplete features shall remain fully functional — only external paste is restricted

3.4.3 Keystroke Session Recorder

- The IDE shall record the complete typing session for every code submission — every keystroke, deletion, pause, and edit
- Instructors shall be able to replay a student's full coding session as a video-like timeline to verify the code was written progressively and not inserted in bulk
- The system shall automatically flag submissions where a large block of code appeared in a single event (indicative of paste via non-standard means or a browser extension)
- Keystroke session data shall be stored securely and retained for the duration of the course

3.4.4 AI-Powered Code Signature Detection

- The system shall integrate an AI model trained to detect patterns characteristic of AI-generated code: unnaturally optimal structure, absence of iteration history, atypical comment density, and naming patterns inconsistent with the student's baseline
- From a student's first week of submissions, the system shall build a coding style baseline profile
- Future submissions shall be compared against this baseline; significant style divergence shall trigger an instructor review flag
- Flagged submissions shall not be automatically penalized — they shall be queued for instructor review

3.4.5 MOSS Code Plagiarism Detection

- All code submissions shall be run through MOSS (Measure of Software Similarity) analysis
- MOSS shall detect structural similarity between submissions from different students even when variable names are changed
- Reports shall be surfaced to instructors with a similarity percentage and side-by-side diff view

3.4.6 Incremental Submission Requirements

- Instructors may configure multi-stage submission gates for larger assignments: e.g., submit after function stubs, submit after core logic, submit after testing
- This mirrors real-world development workflows and makes copying a complete final solution impractical

3.4.7 Code Explain / Viva Mode

- After code submission, the system shall randomly select a configurable percentage of students to complete an async 'Code Explain' challenge
- The student shall record a 2–5 minute screen-recorded video explaining what their code does and the reasoning behind key decisions
- Instructors shall review and score the explanation; a submission with no explanation (when required) shall be held as incomplete
- This feature eliminates AI-assisted cheating since students cannot explain code they did not write

3.4.8 GitHub Repository Integration

- The system shall authenticate with GitHub via OAuth 2.0 using read-only scope
- Instructors shall link specific files or directories from a GitHub repository to a lesson
- Learners shall view repository files with syntax highlighting directly within the lesson view
- Linked file content shall sync hourly with the source repository; commit hash and timestamp shall be displayed
- Write access to any GitHub repository shall never be granted

3.5 Assessment & Proctoring

3.5.1 Standard Quizzes

- Each lesson shall support embedded multiple-choice, true/false, and fill-in-the-blank quizzes
- Quizzes shall support configurable passing thresholds and attempt limits
- Question banks shall be large enough that each student receives a unique randomized set
- Adaptive assessments shall adjust question difficulty in real-time based on the student's prior answers, enabling accurate skill-gap identification

3.5.2 Live Proctored Assessments

- The system shall offer a Proctored Exam mode for high-stakes milestone assessments
- Proctored exams shall require: webcam verification, screen recording, randomized question pools, and time-boxing
- Browser lockdown mode shall prevent new tab creation, copy-paste, window switching, and DevTools access during exams
- A live or async AI proctor shall flag: gaze deviation from screen, secondary person detected in frame, and extended focus loss

- Successful proctored exam completion shall produce a TechLearn Verified Badge — a cryptographically signed credential verifiable by third parties including employers

3.5.3 Biometric Identity Verification

- At the start of all proctored assessments and high-stakes submissions, the system shall perform a face verification check against the student's registration photo
- Biometric data shall be processed locally where possible and shall not be stored beyond the session
- GDPR-compliant consent shall be obtained before biometric features are activated

3.5.4 Timed Scenario Challenges

- Instructors may create timed scenario challenges: a bug, a system design problem, or a code review task presented to the student with a 15–30 minute response window
- The limited time window discourages Googling and rewards genuine preparedness

3.5.5 Applied Project Assessments

- Assessments shall support real-world contextualized project briefs (e.g., 'Build a payroll system for a small business')
- Project briefs shall be configurable per organization, enabling Nigeria/Africa-specific business contexts to be embedded
- Specificity of context makes generic AI-generated solutions ineffective

3.6 Learner Accountability Dashboard

Every student's learning behavior is tracked holistically — not just completion rates, but genuine engagement quality.

3.6.1 Student Learning Health Score

- The system shall compute a composite Learning Health Score for each student, factoring in: video engagement rate, checkpoint pass rate, assignment quality score, quiz performance, time-on-platform consistency, and peer review participation
- The Health Score shall be displayed on the student's own dashboard as a motivational metric and to the instructor dashboard as an early-warning flag

3.6.2 Streak & Consistency Tracking

- The system shall track daily and weekly learning streaks and flag cramming behavior (e.g., 10 hours in one night vs. 1 hour daily)
- Consistency scores shall be weighted in the Learning Health Score to reward sustained engagement

3.6.3 Instructor Accountability View

- Instructors shall see a real-time accountability dashboard showing: who has not logged in this week, who has triggered copy-paste blocks, who failed video checkpoints repeatedly, who passed quizzes but failed the related code exercise, and all flagged submissions
- Instructors shall be able to send a direct nudge message to any at-risk learner from the dashboard

3.6.4 Sponsor / Parent Visibility (Optional)

- For bootcamp or employer-sponsored programs, an optional read-only progress report shall be shareable with designated sponsors or parents
- The report shall show: attendance, module completion, assessment scores, and Health Score — but no personal behavioral flags

3.7 Certifications, Badges & Career Outcomes

3.7.1 Certificates & Badges

- The system shall automatically issue a digital certificate upon course completion
- Certificates shall include: learner name, course title, completion date, instructor name, unique verification ID, and competency breakdown score
- Certificates shall be downloadable as PDF and shareable via a unique public URL
- The system shall issue Open Badges 2.0 compliant digital badges shareable to LinkedIn
- TechLearn Verified Badges (earned via proctored exam) shall carry a cryptographic signature verifiable by employers via a public API endpoint
- Certificates shall display a competency breakdown (e.g., Code Quality: 87%, Problem Solving: 79%) — not just a pass/fail — giving employers meaningful signal

3.7.2 Career Outcome Tracker

- The system shall integrate with LinkedIn and major job boards to track post-certification career outcomes (with user consent)
- Outcome data (e.g., hired, promoted, role change) shall be aggregated at the course level to produce public 'Outcome Stats' — e.g., '74% of graduates who completed this course were hired within 3 months'
- HR Managers shall view outcome data for their employees to calculate training ROI
- This data shall serve as a core sales and marketing differentiator for the platform

3.7.3 Public Portfolio Builder

- All project-based work submitted by a student on the platform shall be automatically compiled into a public portfolio page hosted at [learn.techlearn.io/portfolio/\[username\]](https://learn.techlearn.io/portfolio/[username])
- The portfolio shall showcase: completed projects, earned certificates, competency scores, peer review ratings given and received, and coding activity timeline
- Knowing their work is public to potential employers significantly motivates genuine effort

3.8 Live Sessions / Virtual Classrooms

- The system shall integrate with Zoom API to schedule, launch, and track live sessions
- Session recordings shall be automatically imported and linked to the course module
- Attendance data from Zoom shall be captured and linked to learner progress records
- Students shall receive calendar invites and Slack reminders 24 hours and 1 hour before sessions

3.9 AI Features

3.9.1 AI Pair Programmer / Socratic Tutor

- The IDE shall include a context-aware AI Tutor assistant that can: explain what an error message means, suggest approaches and strategies, answer 'why is my code wrong?', and teach concepts using the student's own code as context
- The AI Tutor shall be explicitly constrained — it can never write code for the student, complete a function, or provide a solution. It operates in explain-only mode
- The AI Tutor shall generate follow-up 'What if?' extension challenges after a student successfully submits code, deepening understanding

3.9.2 AI-Powered Course Recommendations

- The system shall generate personalized course recommendations based on: completed courses, skill assessments, role/profile, and peer behavior patterns
- Recommendations shall update daily and include confidence scores and reasoning labels
- HR Managers shall use AI-generated recommendations to build team learning plans

3.9.3 AI Course Content Generator

- Given a topic and target skill level, the AI generator shall produce: full course outline, module structure, draft quiz question bank (minimum 50 questions per module), and a reading/documentation list sourced from public repos and docs
- Instructors shall review and edit all AI-generated content before publication — AI-generated content shall never be auto-published

3.10 Spaced Repetition Engine

- The system shall implement a Spaced Repetition System (SRS) based on the Ebbinghaus forgetting curve to resurface quiz questions, key concepts, and flashcards at scientifically optimal intervals
- After a student completes a module, the system shall schedule automated micro-review sessions (5–10 minutes each) at 1 day, 3 days, 1 week, 2 weeks, and 1 month intervals
- Micro-review sessions shall be delivered via the platform dashboard and optionally via Slack or email

- Correctly answered items shall have their next review interval extended; incorrectly answered items shall be resurface sooner
- Instructors and HR Managers shall see retention scores alongside completion rates, giving a more accurate picture of actual skill acquisition

3.11 Peer Learning & Social Systems

3.11.1 Peer Code Review

- Students shall be required to complete peer reviews of two classmates' code submissions (using instructor-defined rubrics) before their own submission is accepted
- Peer review teaches code review skills — a critical professional competency — while reinforcing the reviewer's own understanding
- Study groups shall be tracked: if group members submit structurally similar work shortly after a group session, MOSS analysis shall be triggered

3.11.2 Team Hackathon / Challenge Mode

- The system shall support a Cohort Challenge Mode where groups of 3–5 learners collaborate on a time-boxed project inside the platform
- Teams shall have access to: shared multi-user IDE sessions, group video rooms, shared file workspace, and a team submission channel
- Challenges shall include a public leaderboard scored on correctness, code quality, and delivery speed
- Enterprise customers may sponsor internal hackathons using this feature for team skill-building events

3.11.3 Cohort Leaderboards

- Leaderboards shall rank learners on: consistency score, project quality, quiz improvement rate, and peer review contributions — not raw grades
- This rewards growth and effort over prior knowledge, making the leaderboard meaningful for beginners and advanced learners alike

3.12 Instructor Marketplace

- Instructors shall apply to publish on TechLearn through a curated review process — open publishing (Udemy model) shall not be permitted
- All courses shall undergo a quality review by a subject matter reviewer before publication
- Each instructor and course shall receive a continuous Outcome Score based on: learner completion rate, post-quiz score improvement, career outcome data, and learner review quality (not just star rating)
- Top-performing instructors shall receive revenue share boosts, featured placement, and platform promotion
- Low-performing courses shall enter a remediation process before being unpublished

3.13 White-Label Enterprise Portals

- Each enterprise customer shall be provisioned a fully white-labeled subdomain (e.g., learn.acme.com)
- The portal shall support: custom branding (logo, color scheme, typography), a curated course catalog (TechLearn public courses + internal-only private courses), custom certificate templates, and org-specific onboarding flows
- Internal-only courses shall be accessible exclusively to the enterprise's enrolled users
- White-label portals shall have their own admin panel for the enterprise's HR team, with no access to other organizations' data

3.14 Analytics & Reporting

3.14.1 Learner Dashboard

- Students shall see: enrolled courses, completion percentage, quiz scores, time spent, streak, Health Score, and earned badges
- Students shall see their personalized recommended next step

3.14.2 Instructor Analytics

- Instructors shall see: enrollment counts, average completion rate, quiz performance distribution, video checkpoint failure rates, and submission flag counts
- Instructors shall see an Integrity Report per assignment: percentage of flagged submissions, MOSS similarity clusters, Code Explain completion rate
- Instructors shall export analytics reports as CSV or PDF

3.14.3 Admin & HR Dashboard

- Admins shall view platform-wide metrics: total users, active users, courses published, total hours learned
- HR Managers shall view team metrics: employee progress, completion rates, skill coverage, certification status
- HR Managers shall view team Learning Health Scores and retention scores alongside completion rates
- HR Managers shall generate skill-gap reports and training ROI reports informed by career outcome data

3.15 Payment, Billing & Regional Support

- The system shall support Individual, Team, and Enterprise subscription tiers with monthly and annual billing
- Stripe shall be the primary payment processor for global users

- Paystack and Flutterwave shall be integrated as payment processors for Nigerian and African markets
- The system shall support local payment methods common in Nigeria: bank transfer, USSD, and mobile money
- HR Managers shall manage enterprise seat licenses, add/remove seats, and view invoices
- The system shall support promo codes and discount management
- All payment data shall be handled via PCI-DSS compliant third-party processors
- The system shall send automated invoices, renewal reminders, and payment failure notifications

3.16 Offline Mode & Low-Bandwidth Support (PWA)

- TechLearn shall be built as a Progressive Web App (PWA) enabling offline access
- Students shall be able to download video lessons and reading materials for offline viewing
- Downloaded content shall be encrypted with DRM to prevent redistribution
- All offline activity (video progress, notes, quiz attempts) shall be logged locally and synced automatically when connectivity is restored
- Video streaming shall support adaptive bitrate delivery functioning on 2G/3G connections
- The mobile browser IDE shall be optimized for smaller screens to prevent device-based exclusion from coding exercises

3.17 Multi-Language & Localization

- The UI shall support: English, French, Spanish, Arabic, Portuguese, Yoruba, Igbo, Hausa, Mandarin (minimum)
- The system shall support RTL text rendering for Arabic
- Instructors shall upload translated subtitle/caption files (SRT/VTT) for video lessons
- All system emails and notifications shall be sent in the user's preferred language

3.18 Third-Party Integrations

Integration	Purpose	Access Type	v2 Status
Zoom	Live sessions, attendance tracking, recording import	API (OAuth)	Existing
Slack	Notifications, reminders, spaced repetition delivery	Webhook / Bot API	Existing
GitHub	Read course exercise files from repositories	OAuth (read-only)	Existing
Stripe	Global payment processing	API Key	Existing
Paystack	Nigerian & African payment processing	API Key	NEW

Flutterwave	Pan-African payment processing	API Key	NEW
MOSS	Code plagiarism detection	API / CLI	NEW
LinkedIn	Badge sharing, career outcome tracking	OAuth	NEW
Workday / SAP / Salesforce	Enterprise ecosystem data sync	REST API / Webhooks	NEW
Biometric Vendor	Face verification for proctored sessions	API (GDPR compliant)	NEW
AI/ML Engine	Recommendations, code analysis, content generation	Internal / External API	Existing
Email (SendGrid)	Transactional emails, certificates, notifications	SMTP / API	Existing

4. Non-Functional Requirements

4.1 Performance

- Page load time shall not exceed 2 seconds for 95% of requests under normal load
- The platform shall support 50,000+ registered users and 10,000+ concurrent active sessions
- Video streaming shall begin playback within 3 seconds using adaptive bitrate delivery
- The inline IDE shall return code execution results within 5 seconds for standard exercises
- API response time shall not exceed 500ms for 99th percentile requests under load
- Keystroke session recordings shall be streamed and stored asynchronously without impacting IDE responsiveness
- MOSS analysis shall complete within 60 seconds of submission and shall not block the submission acknowledgment

4.2 Scalability

- The backend shall be stateless and horizontally scalable using container orchestration (Kubernetes)
- The database layer shall support read replicas and connection pooling
- Media assets shall be stored on object storage (S3) and delivered via CDN
- Auto-scaling shall trigger based on CPU and memory usage thresholds
- The IDE sandbox runtime pool shall auto-scale independently based on code submission queue depth

4.3 Availability & Reliability

- The platform shall achieve 99.9% uptime (less than 8.7 hours downtime/year)
- The system shall implement automated health checks and failover for all critical services
- Database backups shall occur every 6 hours with RPO of 6 hours
- RTO shall not exceed 1 hour for critical service restoration
- Keystroke recording data shall be replicated to a secondary region to prevent loss

4.4 Security

- All data in transit shall be encrypted using TLS 1.3 or higher
- All data at rest shall be encrypted using AES-256
- The platform shall pass OWASP Top 10 security assessment prior to launch
- MFA shall be enforced for all accounts
- All user-generated inputs shall be sanitized to prevent XSS and SQL injection
- The inline IDE sandbox shall be fully isolated from the host system and network
- Security audit logs shall be retained for a minimum of 12 months

- Keystroke session data shall be stored in an access-controlled, encrypted store accessible only to instructors and admins of the relevant course
- Biometric data shall be processed transiently and never persisted beyond the authentication event

4.5 Compliance

- GDPR compliant: data subject rights (access, deletion, portability) fully supported
- CCPA compliant for California-based users
- WCAG 2.1 AA accessibility standards
- SCORM 1.2 / 2004 and xAPI (Tin Can) standards
- PCI-DSS compliant payment processing
- NDPR (Nigeria Data Protection Regulation) compliant for Nigerian user data
- Explicit GDPR-compliant consent shall be obtained before activating biometric verification, keystroke logging, and webcam proctoring

4.6 Usability

- The UI shall be responsive and optimized for desktop browsers (1280px+ width)
- The mobile browser experience shall be a first-class PWA — not a degraded fallback — with full read/watch mode and a mobile-optimized IDE
- The platform shall achieve a System Usability Scale (SUS) score of 80 or higher
- All interactive elements shall be accessible via keyboard navigation
- Screen readers (NVDA, VoiceOver) shall be supported for learner-facing pages

4.7 Maintainability

- Codebase shall maintain 80%+ unit test coverage for all core modules
- All APIs shall be documented using OpenAPI 3.0 specification
- Feature flags shall enable safe, incremental feature rollouts
- Logging shall be structured (JSON) and aggregated in a centralized log management system
- The public Enterprise REST API and Webhooks system shall be versioned (v1, v2) with a minimum 12-month deprecation notice policy

5. System Architecture Overview

5.1 High-Level Architecture

TechLearn LMS follows a microservices-oriented, event-driven architecture deployed on cloud infrastructure with multi-region support. The key architectural layers are:

- Frontend: Single-Page Application (React) served via CDN, built as a PWA with service workers for offline support
- API Gateway: Rate-limiting, authentication, request routing, and versioning
- Core Microservices: Auth, Course, IDE/Sandbox, Analytics, Notification, Billing, AI Recommendation, Integrity/Proctoring, Spaced Repetition, Portfolio, Marketplace
- Data Layer: PostgreSQL (relational), Redis (cache + session), S3 (media), Elasticsearch (search), InfluxDB (behavioral time-series)
- Event Bus: Kafka for asynchronous inter-service communication (enrollment events, submission events, progress events)
- CDN: CloudFront or equivalent for global video delivery with HLS/DASH adaptive streaming

5.2 Key Service Data Flows

User Action	Services Involved
Login + MFA	Auth Service → RBAC → Device Fingerprint → Session Token
Video Watch	Course Service → CDN → Watch Tracker → Checkpoint Engine → Progress Service
Code Submission	IDE Service → Keystroke Store → Sandbox Runtime → Test Evaluator → MOSS → AI Detector → Progress Service
Proctored Exam	Exam Service → Biometric Verify → Browser Lockdown → AI Proctor → Certificate Service
Course Enrollment	Course Service → Billing Service → Spaced Repetition Scheduler → Notification Service
Spaced Review	SRS Scheduler → Question Bank → Notification → Progress Service
Career Outcome	Certificate Service → LinkedIn API → Outcome Aggregator → HR Dashboard
Portfolio Update	Progress Service → Portfolio Builder → Public CDN

5.3 IDE & Integrity Subsystem Architecture

- Code Execution: Each submission spawns an ephemeral container (10s time limit, 256MB memory, no network access)

- Keystroke Logger: WebSocket-based streaming of keystroke events to a dedicated time-series store; replayed via a purpose-built IDE playback UI
- Copy-Paste Enforcement: CSP headers block clipboard API; JavaScript event listeners intercept paste events before they reach editor state
- MOSS: Submission content is queued to a MOSS worker pool; results returned asynchronously and linked to the submission record
- AI Code Detector: Submission sent to AI analysis microservice; result is a risk score (0–100) and a flag reason; stored against the submission
- Biometric: WebRTC-based face capture is processed by a GDPR-compliant biometric API; match result stored as a boolean + confidence score; raw image is not persisted

6. Security & Compliance

6.1 Authentication Controls

- Passwords hashed using bcrypt with minimum cost factor 12
- Minimum password policy: 12 characters, upper/lower/number/special character
- Session tokens expire after 8 hours of inactivity; refresh tokens max 30 days
- One active session enforced per account at any time

6.2 Academic Integrity Controls Summary

Threat	Control Mechanism
Student pastes external code	Copy-paste block + CSP + paste event logging
Student uses AI to write code	Keystroke recorder + AI signature detection + style baseline
Student copies peer's code	MOSS plagiarism detection + structural similarity analysis
Student doesn't watch videos	Tab focus tracking + active watch detection + seek restriction
Proxy learner (someone else doing work)	Biometric face verification + device fingerprinting + IP tracking
Student can't explain their code	Code Explain / Viva Mode (async video explanation)
Student shares quiz answers	Randomized question banks + adaptive questions
Student skips to end of video	First-watch forward-seek disabled

6.3 Data Privacy

- PII stored in a designated, encrypted database partition
- Users may request full data export (GDPR Article 20) and account deletion (GDPR Article 17)
- Deleted accounts purged within 30 days
- Nigerian users protected under NDPR; a local Data Protection Officer contact shall be published
- Biometric and proctoring data subject to stricter retention limits: maximum 90 days post-course or upon user deletion request

6.4 Security Testing

- Penetration testing before production launch and annually thereafter
- SAST/DAST tools integrated into CI/CD pipeline

- Third-party dependency vulnerabilities scanned weekly; critical CVEs patched within 14 days

7. Appendices

7.1 Extended User Story Summary

ID	Role	User Story	Priority	v2
US-01	Student	As a student, I want to enroll in a course and track my progress holistically via my Health Score.	High	Updated
US-02	Student	As a student, I want to write code in the inline IDE knowing the environment is fair and my work is genuinely mine.	High	Existing
US-03	Student	As a student, I want the AI Tutor to help me understand my errors without giving me the answer.	High	NEW
US-04	Student	As a student, I want to earn a Verified Badge that employers can actually trust.	High	NEW
US-05	Student	As a student, I want to access course material offline when my internet is unreliable.	High	NEW
US-06	Student	As a student, I want my completed projects to build a portfolio visible to potential employers.	Medium	NEW
US-07	Student	As a student, I want spaced repetition to help me retain what I've learned after I complete a course.	High	NEW
US-08	Instructor	As an instructor, I want to replay a student's coding session to verify their work is genuine.	High	NEW
US-09	Instructor	As an instructor, I want MOSS reports to identify when students copy each other's code.	High	NEW
US-10	Instructor	As an instructor, I want to configure video checkpoints so I know students are actually watching.	High	NEW
US-11	Instructor	As an instructor, I want AI to generate a course outline and quiz bank I can refine, not start from scratch.	High	NEW
US-12	Admin	As an admin, I want to manage all users, roles, flags, and review proctored exam integrity reports.	High	Existing
US-13	HR Manager	As an HR Manager, I want to see my team's Learning Health Scores and retention rates, not just completion.	High	NEW
US-14	HR Manager	As an HR Manager, I want career outcome data so I can demonstrate training ROI to leadership.	High	NEW
US-15	HR Manager	As an HR Manager, I want a white-labeled portal with our branding and internal-only courses.	High	NEW

7.2 Competitive Feature Matrix

Feature	TechLearn	Pluralsight	Udemy Biz	Coursera Ent.
Inline IDE with copy-paste block	YES	Partial	No	No
Keystroke session recorder	YES	No	No	No
Code plagiarism detection (MOSS)	YES	No	No	No
Code Explain / Viva Mode	YES	No	No	No
Active video watch detection	YES	No	No	No
Video comprehension checkpoints	YES	No	No	No
Live proctored assessments	YES	No	No	Partial
Cryptographically verified badges	YES	No	No	No
Career outcome tracking	YES	No	No	No
Spaced repetition engine	YES	No	No	No
AI Socratic tutor in IDE	YES	No	No	No
White-label enterprise portals	YES	Yes	Yes	Partial
Team hackathon / challenge mode	YES	No	No	No
Offline PWA mode	YES	No	No	No
Paystack / Flutterwave payment	YES	No	No	No
Public portfolio builder	YES	No	No	No
Enterprise REST API + Webhooks	YES	Partial	No	Partial
Curated instructor marketplace	YES	Yes	No	Yes

7.3 Revision History

Version	Date	Author	Description
1.0	Feb 26, 2026	Product Team	Initial SRS document
2.0	Feb 26, 2026	Product Team	Merged competitive analysis, learner accountability framework, anti-cheating architecture, Nigeria market requirements

7.4 Approval Sign-Off

Role	Name	Signature	Date
Product Manager			
Engineering Lead			
Security Officer			
QA Lead			
Legal / Compliance			